Lecture 1 — Date

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## 1.1 My first section heading

Consider the following nonlinear constrained parametric optimization problem:

$$\begin{aligned} &\min_{x\in C} && f(x,\alpha) \\ &\text{subject to} && \begin{cases} g_i(x,\alpha)=0 & i=1,\ldots,m_1, \\ g_i(x,\alpha)\leqslant 0 & i=m_1+1,\ldots,m, \end{cases} \end{aligned} \tag{1}$$

with the following data:

(a) $C\subset\mathbb{R}^n$ is open, $P$ is a metric space with the metric $d_P$. The functions $C\times P\ni(\xi,\mu)\mapsto f(\xi,\mu)\in\mathbb{R}$ and $C\times P\ni(\xi,\mu)\mapsto g_i(\xi,\mu)\in\mathbb{R}$, $i=1,\ldots,m$ are locally Lipschitz continuous;

(b) for all $\alpha\in P$ the functions $C\ni\xi\mapsto f(\xi,\alpha)\in\mathbb{R}$ and $C\ni\xi\mapsto g_i(\xi,\alpha)\in\mathbb{R}$, $i=1,\ldots,m$ are $\mathscr{C}^2(C;\mathbb{R})$;

(c) for $i=1,\ldots,m$ the maps $\frac{\partial}{\partial x}f(\cdot,\cdot)$ and $\frac{\partial}{\partial x}g_i(\cdot,\cdot)$ are locally Lipschitz from $C\times P$ to $\mathbb{R}^n$, while the maps $\frac{\partial^2}{\partial^2 x}f(\cdot,\cdot)$, $\frac{\partial^2}{\partial^2 x}g_i(\cdot,\cdot)$ are continuous;

(d) $m:=m_1+m_2$, where $m_1$ and $m_2$ are nonnegative integers that corresponds to the dimension of the range-space of $g(x,\alpha)$; i.e., $g(x,\alpha)\in Q\subset\mathbb{R}^m$ implies $Q=\{0\}^{m_1}\times\left(-\mathbb{R}_+^{m_2}\right)$.

Let $\left(x,\alpha\right)\in C\times P$. We define the *active index set* as

$$J(x,\alpha):=\left\{i\in\{m_1+1,\ldots,m\}\;\middle|\;g_i(x,\alpha)=0\right\}. \tag{2}$$

The first order necessary conditions associated with the optimization problem are:

$$\begin{cases} \frac{\partial}{\partial x}f(x,\alpha)+\sum_{i=1}^m\lambda_i\frac{\partial}{\partial x}g_i(x,\alpha)=0, \\ g_i(x,\alpha)=0,\ i=1,\ldots,m_1, \\ g_i(x,\alpha)\leqslant 0,\ \lambda_i\geqslant 0,\ \lambda_i g_i(x,\alpha)=0,\ i=m_1+1,\ldots,m, \end{cases} \tag{3}$$

where the coefficient $\lambda_i$ are the *Lagrange multipliers*. Our primary concern is the pair $\left(x^*,\alpha^*\right)\in C\times P$ such that $x^*$ is a local optimizer of (1) and $x^*(\cdot)$ is Lipschitzian. To this end, here is a result asserting the same:

**Theorem 1.1.** *Consider the multiparametric nonlinear program* (1) *with its associated data* (a)–(d), *and let* $\left(x^*,\alpha^*,\lambda^*\right)\in C\times P\times\mathbb{R}^m$ *such that:*

- *the vectors* $\frac{\partial}{\partial x}g_i(x^*,\alpha^*)$, $i\in\{1,\ldots,m_1\}\cup J(x^*,\alpha^*)$ *are linearly independent;*
- *for all* $0\neq b\in\mathbb{R}^n$ *such that* $\left\langle\frac{\partial}{\partial x}f(x^*,\alpha^*),b\right\rangle=0$ *and* $\left\langle\frac{\partial}{\partial x}g_i(x^*,\alpha^*),b\right\rangle=0$, $i\in\{1,\ldots,m_1\}\cup\{i\in J(x^*,\alpha^*)\mid\lambda_i^*>0\}$, *the following hold:*

$$\left\langle\left(\frac{\partial^2}{\partial^2 x}f(x^*,\alpha^*)+\sum_{i=1}^m\lambda_i^*\frac{\partial^2}{\partial^2 x}g_i(x^*,\alpha^*)\right)b,b\right\rangle>0.$$

*If $\left(x^*, \alpha^*, \lambda^*\right)$ satisfies the first-order conditions (3), then there exist neighbourhoods $C_0, V_0$ of $x^*$ and $\alpha^*$ in $C$ and in $P$, respectively and mappings $x : V_0 \longrightarrow C_0$, $\lambda : V_0 \longrightarrow \mathbb{R}^m$ such that $x(\cdot)$ and $\lambda(\cdot)$ are Lipschitzian and $x(\alpha^*) = x^*$, and $\lambda(\alpha^*) = \lambda^*$.*

**Proof:** For a proof we refer the readers to [1, Corollary 2.3]. □

### 1.1.1 A subsection heading

Regarding macros: most of the mathematical symbols like sets, functions, operations etc are all defined in the sty file. Please use them, for example do not write:

$$\mathbb{R}^d, \mathbb{Q}, \mathscr{C}^2, C, x, \min, \text{subject to}, e^{2x^2}, \|x\|, \langle x, y\rangle, f : X \to Y, \nabla, \nabla^2$$

All of these are already predefined in the sty file, use these instead:

$$\mathbb{R}^d, \mathbb{Q}, \mathscr{C}^2, C, x, \min, \text{subject to}, \mathrm{e}^{2x^2}, \left\|x\right\|, \left\langle x,\, y\right\rangle, f : X \to Y, \nabla, \nabla^2$$

Here is how to typeset an array of equations.

$$x = y + z \tag{4}$$

$$\alpha = \frac{\beta}{\gamma} \tag{5}$$

And a table.

| **Method** | Cost | Iterations |
|---|---|---|
| Naive descent | 12 | 200 |
| Newton's method | 500 | 30 |

**Table 1.1.** Comparison of different methods.

### 1.1.2 Yet another subsection

**Corollary 1.2.** *A corollary of Theorem 1.1.* □

**Algorithm 1:** Some Algorithm

**Data** : $u_0$
**Initialize:** Lipschitz constant $L$ of $u_0$
**while** *the function* $u_0 \in \mathbb{S}$ **do**
  Do something
  continue
**end**
**else if** $u_0 \notin \mathbb{S}$ **then**
  Do something else.
**end**
**Output** : job done!

# Bibliography

[1] B. Cornet and J.-PH. Vial. Lipschitzian solutions of perturbed nonlinear programming problems. *SIAM Journal on Control and Optimization*, 24(6):1123–1137, 1986.